

UNIT 1

Step-wise Thinking



Learning Objectives

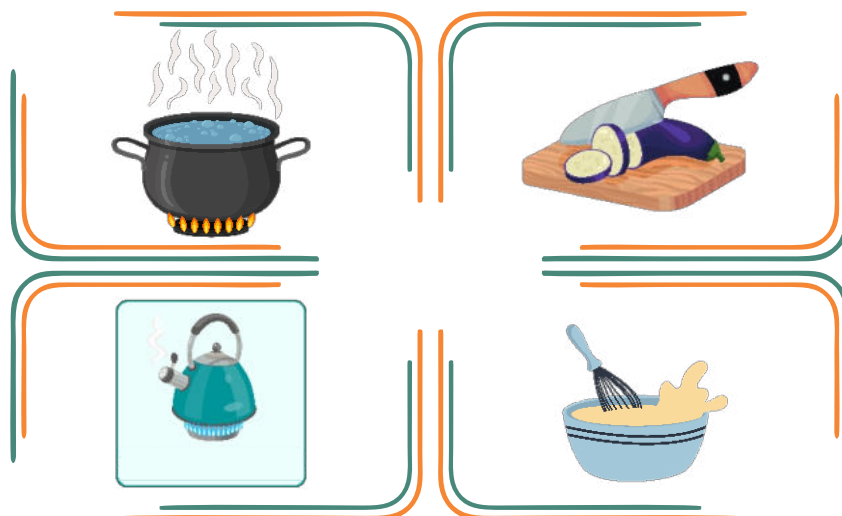
- Understand how to break down activities into smaller, manageable steps.
- Learn the importance of sequencing in problem-solving and programming.
- Write step-wise instructions to solve a problem or perform a task.

Step-wise thinking means breaking down a big task into smaller, easy steps, just like following instructions in a game or a recipe. By doing each step in the right order, we can complete the task without missing anything. This is how computers work too! They follow small steps called instructions to solve problems and complete tasks. In programming, these steps are called an algorithm.



Why Step-wise Thinking is Important

- 1. Easier to Understand:** Breaking down a big problem helps us understand each part better.
- 2. Clear Instructions:** When we do things step-by-step, it's easy to see what comes next.
- 3. No Mistakes:** Following steps in order reduces mistakes, just like adding ingredients one at a time when cooking.



Example of Step-wise Thinking in Real Life

Imagine we want to make a cup of tea. Here's how we'd break down the steps:

1



Boil water in a kettle.

2



Place a tea bag in a cup.

3



Pour the hot water into the cup.

4



Let the tea bag steep for a few minutes.

5



Add sugar or milk if you like.

6



Stir and enjoy your tea!

Learning to Think Like a Computer

Computers need exact instructions to work correctly. If we don't tell them exactly what to do, they won't understand the task. Let's look at how we'd teach a computer to do something using step-wise thinking.

Practical Example: Organizing Books on a Shelf

Step - by - Step Instructions for a Computer:

1. Pick up the first book.
2. Place it on the left side of the shelf.
3. Pick up the next book.
4. Check if it's taller than the book already on the shelf.
5. If yes, place it on the right; if no, place it on the left.
6. Repeat until all books are placed.



Know the Key Concepts and Terms

- 1. Step-wise Thinking:** Breaking down a big task into small, clear steps.
- 2. Algorithm:** A list of steps to solve a problem or complete a task.
- 3. Sequencing:** Putting steps in the correct order to avoid mistakes.
- 4. Programmers:** A programmer is someone who writes instructions to tell computers what to do. They use special computer languages to create programs for games, apps and other software.

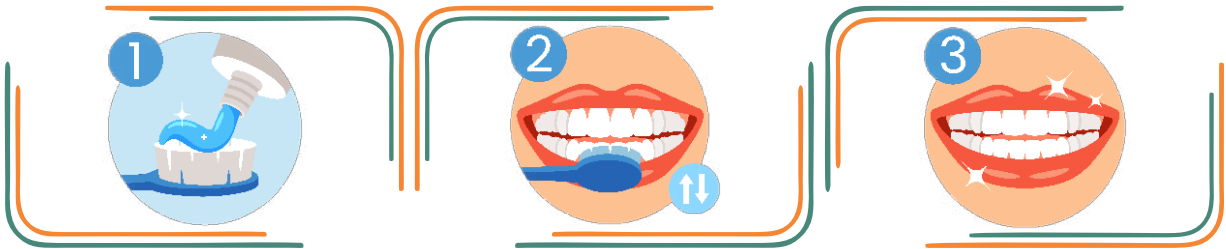
EXERCISES

01: Match the terms with their correct definitions.

Term
Algorithm
Sequencing
Step-wise Thinking
Computer

Definition
A machine that follows instructions to complete tasks
A list of steps to complete a task
Breaking down a big task into smaller steps
Putting steps in the correct order

02: Think about something you know how to do well, like drawing a picture or brushing your teeth. Write down each small step you would follow to complete the task. Remember, even small actions should be included!

[illegible]

03: Fill in the blanks.

1. Stepwise thinking means breaking a _____ into smaller steps.
2. Computers need _____ to complete tasks.
3. A set of instructions for a task is called an _____.
4. Following steps in order reduces _____.
5. To make a cup of tea, the first step is to _____ water.

04: Write True or False for each statement.

	TRUE	FALSE
1. Step-wise thinking makes tasks harder to complete.	☐	☐
2. An algorithm is a list of steps to complete a task.	☐	☐
3. Computers understand big tasks without any steps.	☐	☐
4. Step-wise thinking is only useful for cooking.	☐	☐
5. Each step in an algorithm should be in the correct order.	☐	☐

05: Answer the following questions.

1. What is step-wise thinking, and why is it important?
2. What is an algorithm?
3. What could happen if we do not follow the correct sequence in step-wise thinking?
4. Why do computers need instructions in small steps to perform a task?
5. Choose one of the following tasks and write the steps:
 - Making a glass of lemonade
 - Setting the table for dinner
 - Drawing a house

[illegible]

Lined area for writing.

DRAWING A SIMPLE HOUSE IN MS PAINT

- 1. Open MS Paint:** Click on the MS Paint application to open it.
- 2. Select a shape tool:** Choose the rectangle tool to draw the base of the house.
- 3. Draw the house base:** Click and drag to create a rectangle for the house.
- 4. Draw the roof:** Select the triangle shape and place it on top of the rectangle for the roof.
- 5. Add windows:** Use the rectangle tool again to draw smaller rectangles for the windows.
- 6. Draw the door:** Select the rectangle tool to draw a door in the base of the house.
- 7. Add colours:** Use the paint bucket tool to fill in colours for the house, roof, windows, and door.
- 8. Save the image:** After finishing, click on "File" and then "Save" to save your work.

